

Lesson 03 Demo 01

Creating a Secure EKS Cluster

Objective: To create a private Amazon EKS cluster using the AWS Management Console, including setting up the VPC, subnets, IAM roles, and security settings, to ensure secure and isolated Kubernetes workloads

Tools required: AWS Console

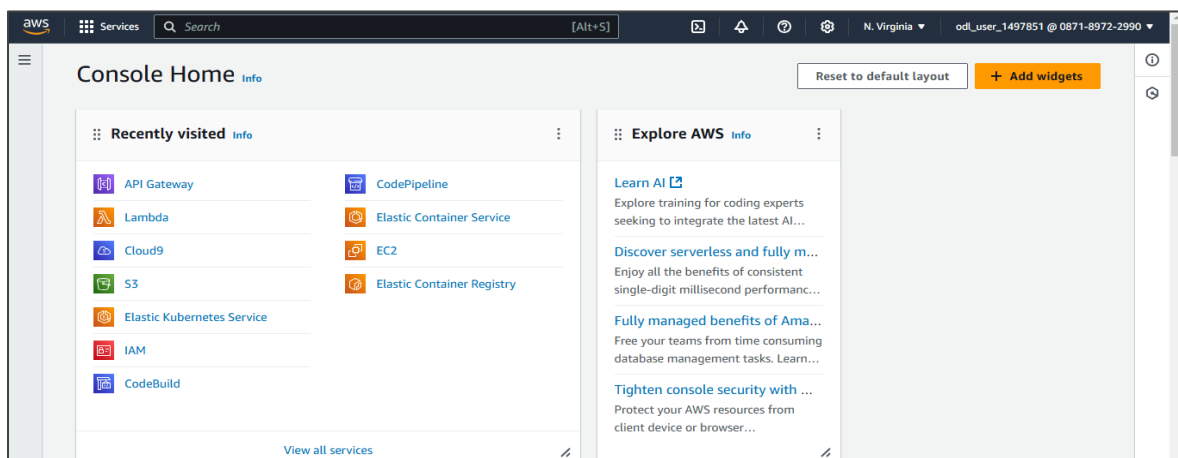
Prerequisites: None

Steps to be followed:

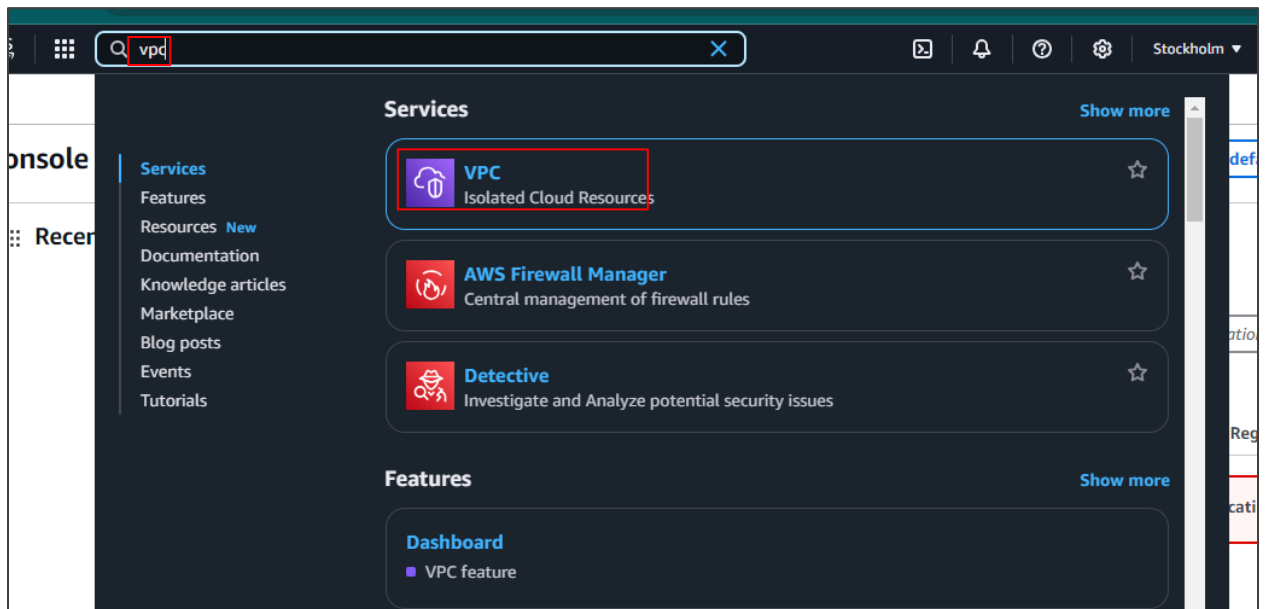
1. Create a VPC
2. Configure secure EKS Cluster with private subnets

Step 1: Create a VPC

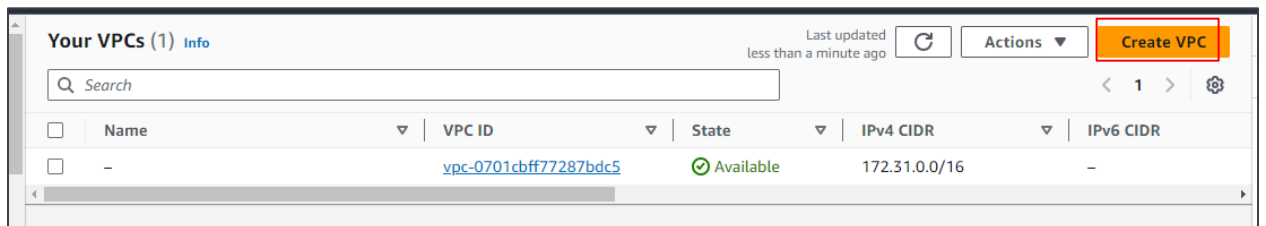
1.1 Open the **AWS Management Console** in your practice environment



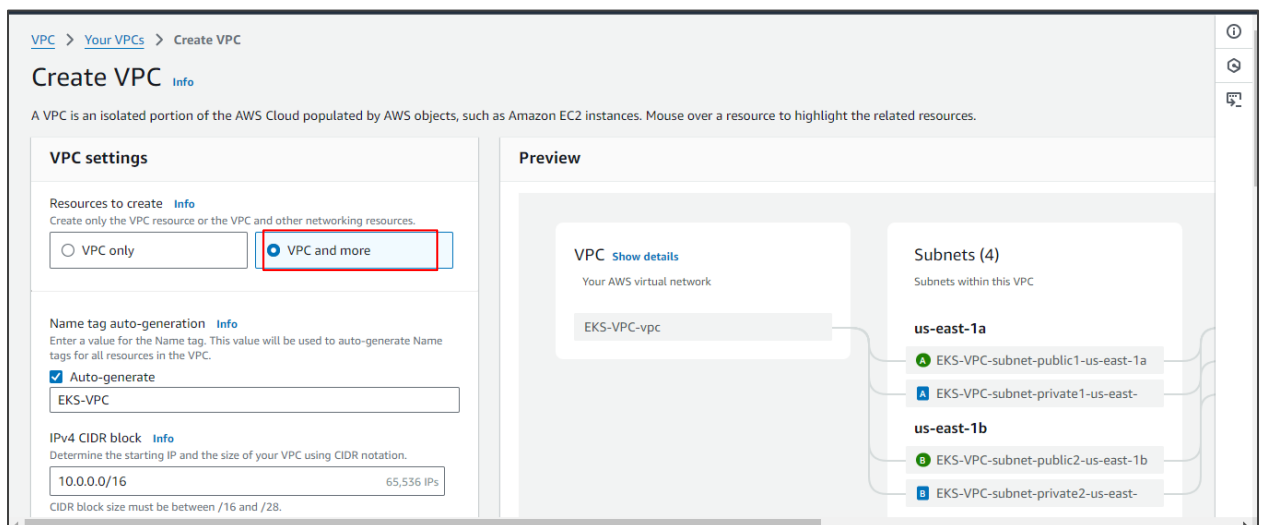
1.2 In the search bar, type **VPC** and select it



1.3 Click on **Create VPC**



1.4 Select the **VPC settings** as **VPC and more** and keep the default values for other settings, or adjust them as needed



Tenancy [Info](#)

Default

Number of Availability Zones (AZs) [Info](#)

Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1

2

3

► Customize AZs

Number of public subnets [Info](#)

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0

2

Number of private subnets [Info](#)

The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0

2

4

► Customize subnets CIDR blocks

1.5 Click on **Create VPC**

► Customize subnets CIDR blocks

NAT gateways (\$) [Info](#)

Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway

None

In 1 AZ

1 per AZ

VPC endpoints [Info](#)

Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None

S3 Gateway

DNS options [Info](#)

☒ Enable DNS hostnames

☒ Enable DNS resolution

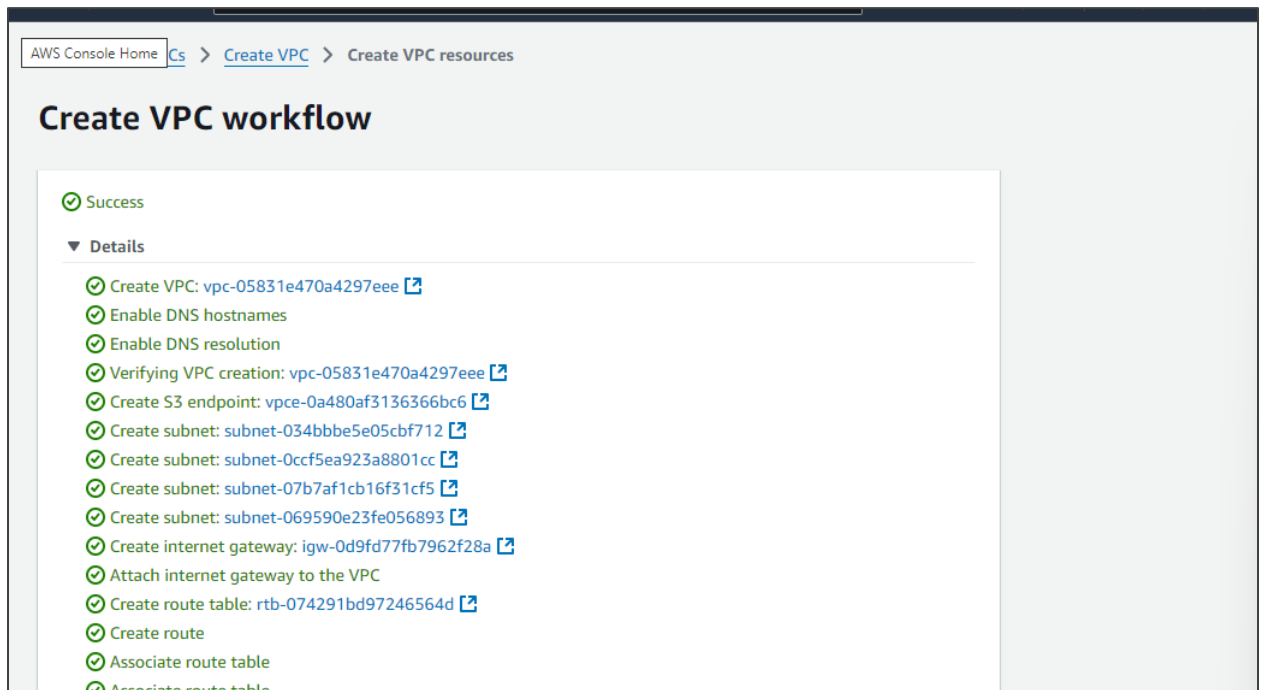
► Additional tags

Cancel

 Preview code

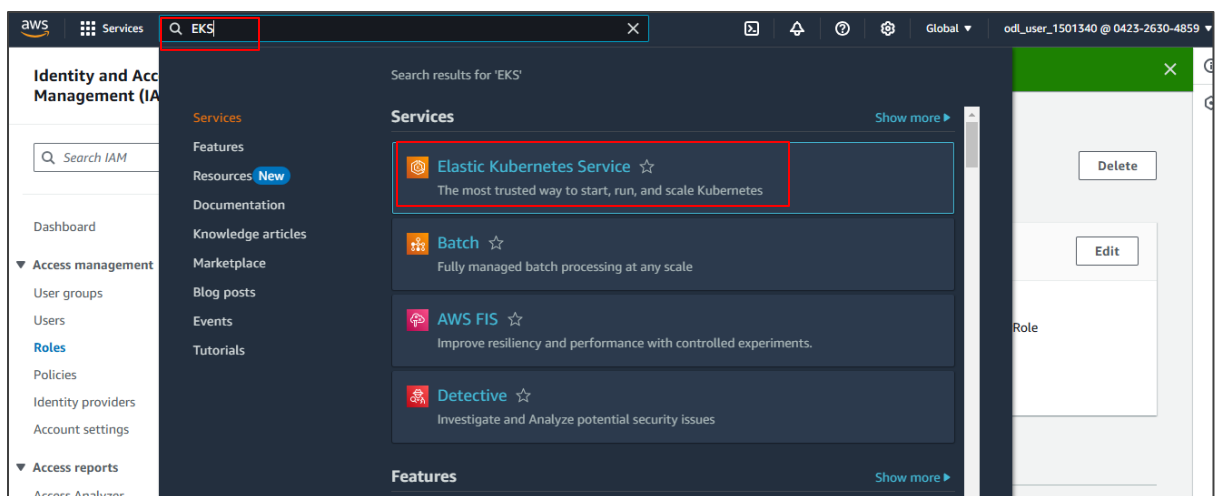
Create VPC

You will see the below screen once created.

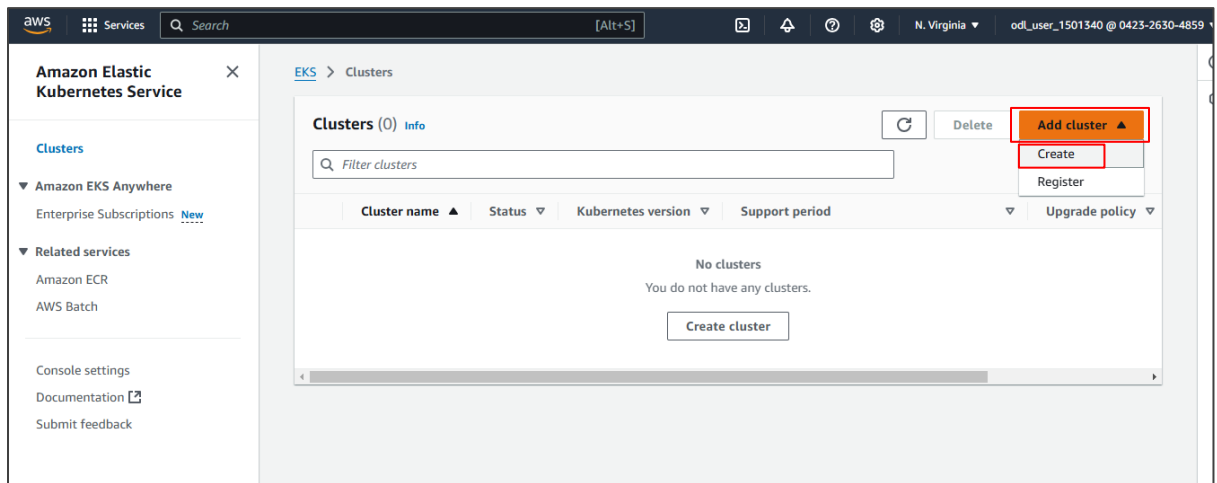


Step 2: Configure secure EKS Cluster with private subnets

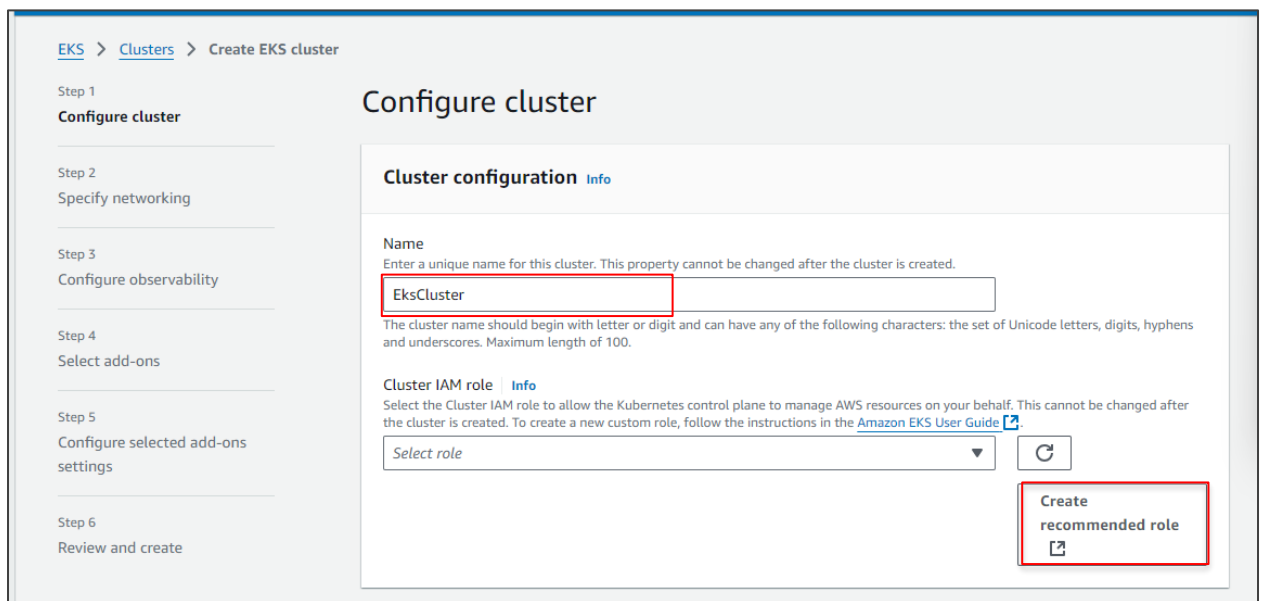
2.1 In the AWS Management Console, search for EKS and select Elastic Kubernetes Service



2.2 Click on **Add cluster** and then select **Create**



2.3 Provide a desired name for your EKS cluster and click on **Create recommended role** to create an IAM role



2.4 Select **AWS service** from the **Trusted entity type** section

The screenshot shows the AWS IAM console 'Create role' page. The breadcrumb navigation is 'IAM > Roles > Create role'. On the left, a progress bar indicates three steps: 'Step 1: Select trusted entity' (active), 'Step 2: Add permissions', and 'Step 3: Name, review, and create'. The main heading is 'Select trusted entity' with an 'Info' link. Below this, the 'Trusted entity type' section contains five radio button options: 'AWS service' (selected and highlighted with a red box), 'AWS account', 'Web identity', 'SAML 2.0 federation', and 'Custom trust policy'. Each option includes a brief description of its function.

Step 1
● **Select trusted entity** Info

Step 2
○ Add permissions

Step 3
○ Name, review, and create

Select trusted entity

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

2.5 Select **EKS-Cluster** as **Use case** and click **Next**

The screenshot shows the AWS IAM console 'Create role' page, Step 2: Add permissions. The breadcrumb navigation is 'IAM > Roles > Create role'. The main heading is 'Add permissions'. Below this, the 'Use case' section is active. It includes a description: 'Allow an AWS service like EC2, Lambda, or others to perform actions in this account.' and a 'Service or use case' dropdown menu with 'EKS' selected. Below the dropdown, a message says 'Choose a use case for the specified service.' followed by a list of use cases. The 'EKS - Cluster' option is selected and highlighted with a red box. Other use cases include 'EKS - Service', 'EKS - Nodegroup', 'EKS - Fargate pod', 'EKS - Fargate profile', 'EKS - Connector', and 'EKS Local - Outpost'.

Step 1
● **Select trusted entity** Info

Step 2
○ **Add permissions**

Step 3
○ Name, review, and create

Add permissions

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EKS ▼

Choose a use case for the specified service.

Use case

- ☐ **EKS - Service**
Allows EKS to manage clusters on your behalf.
- ☒ **EKS - Cluster**
Allows the cluster Kubernetes control plane to manage AWS resources on your behalf.
- ☐ **EKS - Nodegroup**
Allow EKS to manage nodegroups on your behalf.
- ☐ **EKS - Fargate pod**
Allows access to other AWS service resources that are required to run Amazon EKS pods on AWS Fargate.
- ☐ **EKS - Fargate profile**
Allows EKS to run Fargate tasks.
- ☐ **EKS - Connector**
Allows access to other AWS service resources that are required to connect to external clusters.
- ☐ **EKS Local - Outpost**
Allows Amazon EKS Local to call AWS services on your behalf.

☐ EKS - Fargate pod
Allows access to other AWS service resources that are required to run Amazon EKS pods on AWS Fargate.

☐ EKS - Fargate profile
Allows EKS to run Fargate tasks.

☐ EKS - Connector
Allows access to other AWS service resources that are required to connect to external clusters

☐ EKS Local - Outpost
Allows Amazon EKS Local to call AWS services on your behalf.

☐ EKS - Pod Identity
Allows pods running in Amazon EKS cluster to access AWS resources.

Cancel

Next

2.6 Enter the Role name

IAM > Roles > Create role

Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

EksRole

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

Description

Add a short explanation for this role.

Allows the cluster Kubernetes control plane to manage AWS resources on your behalf.

2.7 Now, click on Create role

ate role

Step 2: Add permissions

Edit

Permissions policy summary

Policy name	Type	Attached as
AmazonEKSClusterPolicy	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel

Previous

Create role

2.8 Navigate back to the cluster and select the newly created IAM role

The screenshot shows the AWS Management Console interface for creating an EKS cluster. The breadcrumb navigation at the top reads 'EKS > Clusters > Create EKS cluster'. On the left, a sidebar lists six steps: 'Step 1: Configure cluster', 'Step 2: Specify networking', 'Step 3: Configure observability', 'Step 4: Select add-ons', 'Step 5: Configure selected add-ons settings', and 'Step 6: Review and create'. The main content area is titled 'Configure cluster' and contains a 'Cluster configuration' section. Within this section, the 'Name' field is set to 'EksCluster'. The 'Cluster IAM role' dropdown menu is set to 'EksRole' and is highlighted with a red rectangle. To the right of the dropdown is a 'Create recommended role' button. The top of the console shows the AWS logo, 'Services' menu, a search bar, and the user's location 'N. Virginia' and username 'odl_user_1519'.

2.9 Click Next

This screenshot shows the 'Tags' section of the AWS Management Console for the EKS cluster. At the top, there is a notification about ARC zonal shift, with a 'Learn more' link. Below this, the 'Tags (0)' section indicates that no tags are currently associated with the resource. There is an 'Add new tag' button and a note stating 'You can add up to 50 tags.' At the bottom right of the console, there are two buttons: 'Cancel' and 'Next'. The 'Next' button is highlighted with a red rectangle, indicating the next step in the process.

2.10 Select the previously created **VPC** and two **Subnets**. Then, select the default **Security groups**.

EKS > Clusters > Create EKS cluster

Step 1
[Configure cluster](#)

Step 2
Specify networking

Step 3
[Configure observability](#)

Step 4
[Select add-ons](#)

Step 5
[Configure selected add-ons settings](#)

Step 6
[Review and create](#)

Specify networking

Networking [Info](#)
IP address family and service IP address range cannot be changed after cluster creation.

VPC [Info](#)
Select a VPC to use for your EKS cluster resources.

vpc-05831e470a4297eee | EKS-VPC-vpc

Subnets [Info](#)
Choose the subnets in your VPC where the control plane may place elastic network interfaces (ENIs) to facilitate communication with your cluster. To create a new subnet, go to the corresponding page in the [VPC console](#).

Select subnets

subnet-0ccf5ea923a8801cc | EKS-VPC-subnet-public2-us-east-1b
us-east-1b 10.0.16.0/20

subnet-034bbbe5e05cbf712 | EKS-VPC-subnet-public1-us-east-1a
us-east-1a 10.0.0.0/20

Clear selected subnets

Security groups [Info](#)

Step 6
[Review and create](#)

us-east-1b 10.0.16.0/20

subnet-034bbbe5e05cbf712 | EKS-VPC-subnet-public1-us-east-1a
us-east-1a 10.0.0.0/20

Security groups [Info](#)
Choose the security groups to apply to the EKS-managed Elastic Network Interfaces that are created in your control plane subnets. To create a new security group, go to the corresponding page in the [VPC console](#).

Select security groups

sg-0b244a782ab09d6d4 | default
default VPC security group

Clear selected security groups

Choose cluster IP address family [Info](#)
Specify the IP address type for pods and services in your cluster.

☒ IPv4

☐ IPv6

Configure Kubernetes service IP address block [Info](#)

☐ Specify the range from which cluster services will receive IP addresses.

2.11 Select **Cluster endpoint access** as **Private** and click **Next**

Cluster endpoint access [Info](#)
Configure access to the Kubernetes API server endpoint.

☐ **Public**
The cluster endpoint is accessible from outside of your VPC. Worker node traffic will leave your VPC to connect to the endpoint.

☐ **Public and private**
The cluster endpoint is accessible from outside of your VPC. Worker node traffic to the endpoint will stay within your VPC.

☒ **Private**
The cluster endpoint is only accessible through your VPC. Worker node traffic to the endpoint will stay within your VPC.

Cancel Previous **Next**

2.12 Enable the Control plane logs you need and click **Next**

Control plane logs [Info](#)
Send audit and diagnostic logs from the Amazon EKS control plane to CloudWatch Logs.

☒ **API server**
Logs pertaining to API requests to the cluster.

☒ **Audit**
Logs pertaining to cluster access via the Kubernetes API.

☒ **Authenticator**
Logs pertaining to authentication requests into the cluster.

☒ **Controller manager**
Logs pertaining to state of cluster controllers.

☒ **Scheduler**
Logs pertaining to scheduling decisions.

Cancel Previous **Next**

2.13 Click **Next** to proceed through the configuration screens

Gain visibility into your Kubernetes resources to optimize your spend and allocate more effectively [Product details](#)

Category	Listed by	Supported versions	Pricing starting at
cost-management	Kubecost	1.29, 1.28, 1.27, 1.26, 1.25, 1.24, 1.23, 1.22, 1.21, 1.20	View pricing details

**kubearmor**
KubeArmor offers robust Runtime protection for Kubernetes & other cloud workloads. Leveraging eBPF and Linux Security Modules (LSM), it enforces policy-based security measures. Safeguard your cloud native applications by limiting any suspicious behavior effectively across diversified workloads. Abstract away complexity, and embrace security. [Product details](#)

Standard Contract

Category	Listed by	Supported versions	Pricing starting at
security	AccuKnox	1.27, 1.26, 1.25, 1.24, 1.23	View pricing details

Cancel Previous **Next**

2.14 Review everything and click **Create**

[EKS](#) > [Clusters](#) > Create EKS cluster

Step 1
[Configure cluster](#)

Step 2
[Specify networking](#)

Step 3
[Configure observability](#)

Step 4
[Select add-ons](#)

Step 5
[Configure selected add-ons settings](#)

Step 6
Review and create

Review and create

Step 1: Cluster Edit

Cluster configuration

Name	Kubernetes version
EksCluster	1.31
Upgrade policy	Cluster IAM role
Standard	arn:aws:iam::471112684403:role/EksRole
Kubernetes cluster administrator access	Authentication mode
Allow cluster administrator access	EKS API

ARC Zonal shift

< 1 >

Add-on name ▲	Version ▼
coredns	v1.11.3-eksbuild.1
eks-pod-identity-agent	v1.3.2-eksbuild.2
kube-proxy	v1.31.0-eksbuild.2
vpc-cni	v1.18.3-eksbuild.2

EKS Pod Identity (1) < 1 >

Add-on name ▲	IAM role 🔗 ▼	Service account ▼
vpc-cni	Not set	aws-node

[Cancel](#)
[Previous](#)
[Create](#)

Once created, you can see that the EKS Cluster is launched as shown below:

EKS > Clusters > EKSCluster

[🔄](#)
[Delete cluster](#)

EKSCluster

▼ Cluster info [Info](#)

Status 🟢 Active	Kubernetes version Info 1.31	Support period 🔗 Standard support until November 26, 2025	Provider EKS
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[Overview](#)
[Resources](#)
[Compute](#)
[Networking](#)
[Add-ons](#)
[Access](#)
[Observability](#)
[Upgrade insights](#)
[Update history](#)
[Tags](#)

Details

API server endpoint https://2C60DD2FAB64D3134A7DDEDE526D6993.gr7.us-east-1.eks.amazonaws.com	OpenID Connect provider URL https://oidc.eks.us-east-1.amazonaws.com/id/2C60DD2FAB64D3134A7DDEDE526D6993	Created 12 minutes ago
Certificate authority LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSB0tLS0tCk1JSURCVENDQWUyZ0F3SUJBZ0lWXBil2crRVowVF13RFFZSkVWklodmNOQVFFTEJRQXdGVEVUTUJFR0E	Cluster IAM role ARN arn:aws:iam::013845847727:role/EKSRole View in IAM	Cluster ARN arn:aws:eks:us-east-1:013845847727:cluster/EKSCluster
Platform version Info eks.12		

By following these steps, you have successfully created a private Amazon EKS cluster using the AWS Management Console. This includes setting up the VPC, subnets, IAM roles, and security settings to ensure secure and isolated Kubernetes workloads.